

# Deepak Mallampati

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Kent, OH -- F-1 OPT, open to sponsorship

## PROFESSIONAL SUMMARY

Applied AI / LLM engineer (2.5+ years) shipping production multi-agent RAG and fine-tuned models with measurable retrieval and hallucination gains. Track record: RAG (~35% retrieval precision gain), agentic LLM workflows (>30% hallucination reduction), and structured prompt transformation layers (+30% contextual alignment in multimodal pipelines). Strong self-starter ship discipline -- independently shipped **Manga Lens** (live Chrome extension, 4 AI providers across 29 sites) and **Agentic Healthcare Claims** pipeline with schema-validated JSON contracts. Comfortable owning the full loop: prompt optimization, fine-tuning, multi-agent orchestration, evaluation, observability.

## CORE COMPETENCIES

LLM Engineering & Fine-Tuning . Prompt Optimization . Multi-Agent RAG . LangChain / LlamaIndex . Python (3+y effective) . Evaluation Pipelines & Observability . Structured Outputs & Guardrails . EDA & Pragmatic Modeling . Self-Starter / Project Ownership . Production AI Deployment . Multimodal Systems

## EXPERIENCE

### Progress Solutions Inc.

Jul 2025 -- Present

Jr. AI/ML Engineer Trainee -- USA | AI Consulting, Production LLM Systems

- Built and deployed **multi-agent RAG systems** with recursive semantic chunking + transformer embeddings; ~35% retrieval precision gain and >30% irrelevant-result reduction; grounded-response alignment exceeded 90% in evaluation.
- Developed **agentic LLM workflows** with structured reasoning, tool discipline, grounding rules, and inter-step evaluation; improved agent response stability by ~25%; reduced hallucination rate >30%.
- Built evaluation + observability for LLM services: structured logging, load simulation, audit trails; reduced post-deployment defects by ~30%.
- Packaged ML/LLM inference as FastAPI/Flask services containerized with Docker; full lineage + system-limitation documentation for stakeholders.
- Ran EDA on multi-source datasets (Pandas/NumPy); raised dataset reliability above 98% and cut downstream model instability by ~40%.

### Suvidha Foundation

Jan 2022 -- Mar 2022

Machine Learning Engineer -- Hyderabad, India | Nonprofit Tech, Video Analytics, Applied NLP

- Built **transformer-based video summarization** for 5,000+ sessions; hierarchical summarization + timestamp-aligned clip extraction reduced manual review time 60-70%; ~85% alignment with human curation.
- Implemented **document Q&A with RAG**; reduced hallucinated outputs ~30% and raised contextual answer accuracy >85%.

### Energy Solutions International (Emerson)

Jun 2022 -- Apr 2023

Database & DevOps Performance Engineer (Intern) -- Hyderabad, India

- Built CI/CD pipelines with Jenkins; reduced deployment errors >30%; improved release cycle ~35-40%.
- Designed Grafana observability dashboards for production SQL systems; cut incident recurrence ~25%.

## PROJECTS

### Manga Lens

shipped -- Chrome Web Store

Solo-built and shipped Chrome extension integrating **4 AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v local). Manifest V3 + content scripts + service workers; multi-section capture pipeline with viewport-slice screenshots and coordinate remapping; 7-day translation cache; per-domain configs for 29 sites. Full ownership: design, code, Chrome review, user feedback loop.

TypeScript, Chrome APIs, Manifest V3, Claude/GPT-4o/Ollama, provider-agnostic abstraction

## Agentic Healthcare Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline (Intake -> Validation -> Consistency -> Duplicate Detection -> Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation; ANN similarity for duplicates; explainable risk scoring with audit-ready reasoning traces.

*Python, LangChain, vector search, FastAPI, Docker*

## Dream Decoder -- AI Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript app with coordinated multimodal APIs (Perplexity Sonar, Replicate image models). Introduced **intermediate structured prompt transformation layers** -- improved contextual alignment ~**30%** and first-pass image success rate ~**25-30%** over naive direct prompting.

*FastAPI, React, TypeScript, Perplexity Sonar, Replicate*

## Agentic Pixel Character Synthesis Engine

Phased GenAI system for identity-consistent, pose-controlled pixel character synthesis. **Stable Diffusion fine-tuning + LoRA** for identity preservation; ControlNet + OpenPose/MediaPipe for pose conditioning; **LLM-based agent orchestrator** that decomposes high-level prompts into generation tasks.

*PyTorch, HF Diffusers, FastAPI, Docker, LoRA, ControlNet*

## EDUCATION

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**Master's Degree** -- Kent State University, Ohio, USA

*Aug 2023 -- May 2025*

## TECHNICAL SKILLS

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**Languages & Frameworks:** Python, TypeScript, JavaScript, React, FastAPI, Flask, SQL (PostgreSQL, T-SQL)

**LLM & GenAI:** LangChain, LlamaIndex, RAG, multi-agent workflows, prompt engineering, fine-tuning, evaluation pipelines, structured outputs, guardrails, grounding

**ML Libraries:** PyTorch, Hugging Face Transformers, scikit-learn, XGBoost, Diffusers, ControlNet, LoRA

**Developer Tooling:** Cursor, Claude Code, Docker, CI/CD (Jenkins), Grafana, RESTful APIs, MLOps/LLMOps

**Work Authorization:** F-1 OPT -- authorized to work in the United States now; future sponsorship required